

A PROJECT REPORT

on

**House Price Prediction with Reliable Price Ranges
and Clear Explanations**

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BACHELOR OF TECHNOLOGY

IN

COMPUTER SCIENCE AND ENGINEERING

BY

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UNDER THE GUIDANCE OF

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SCHOOL OF COMPUTER ENGINEERING

KIIT DEEMED TO BE UNIVERSITY

BHUBANESWAR, ODISHA – 751024

AUGUST 2026

**KIIT Deemed to be University
School of Computer Engineering
Bhubaneswar, ODISHA – 751024**



CERTIFICATE

This is to certify that the project report entitled “**House Price Prediction with Reliable Price Ranges and Clear Explanations**” submitted by **Deepjyoti Bhattacharyya (Roll No. 23053381)** to KIIT Deemed to be University, Bhubaneswar, in partial fulfilment of the requirement for the award of the degree of **Bachelor of Technology in Computer Science and Engineering**, is a bona fide record of work carried out by him under my supervision and guidance.

To the best of my knowledge, the matter embodied in this report has not been submitted elsewhere for the award of any other degree.

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Signature of Guide

Place: Bhubaneswar

Ms. Monalisha Mahapatra

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School of Computer Engineering

KIIT Deemed to be University

DECLARATION

I hereby declare that the project report entitled “**House Price Prediction with Reliable Price Ranges and Clear Explanations**” submitted by me to the School of Computer Engineering, KIIT Deemed to be University, is a record of bona fide work carried out by me under the supervision of Ms. Monalisha Mahapatra.

I further declare that the work reported in this project has not been submitted, either in part or in full, for the award of any other degree or diploma of this institute or of any other institute or university, and that all datasets, libraries, and prior published work used are duly acknowledged and cited in this report.

Date:

Place: Bhubaneswar

Deepjyoti Bhattacharyya

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I am also grateful to the School of Computer Engineering, KIIT Deemed to be University, for providing the academic environment, coursework, and computing resources that made this project possible.

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Deepjyoti Bhattacharyya

ABSTRACT

Most house-price tools, including commercial ones, report a single confident-looking number and a single average accuracy figure, giving a user no way to tell whether their specific property is one the model handles well. Zillow's 2021 closure of its own home-buying business, after its point estimates proved far less reliable than advertised, is a documented real-world consequence of this gap. This project builds a price estimator for residential flats in six Indian metro cities (Bangalore, Chennai, Delhi, Hyderabad, Kolkata, Mumbai) that reports a predicted price, a 90% price range whose empirical reliability is measured rather than assumed, a per-house explanation of which features drove the estimate, and an explicit low-confidence flag for houses unlike the training data. Ridge regression, Random Forest, and XGBoost were tuned by cross-validation on a training split; XGBoost was selected and its price ranges were built with conformal prediction (MAPIE), using both split-conformal and an adaptive, quantile-based variant, calibrated on a held-out split never used for fitting. On a test split touched exactly once, all four range-construction methods achieved coverage within one point of the 90% target. Point accuracy was 51.1% MAPE, a genuine limitation of a dataset built from unverified asking prices with no transaction date and roughly 70% missing amenity data, reported honestly rather than averaged away. Predictions flagged low-confidence showed 62.1% MAPE versus 50.4% for the rest, confirming the flag finds real weaknesses. The system is deployed as a FastAPI service with a hand-built frontend, backed by a Streamlit fallback, both serving the same pipeline.

Keywords: house price prediction, conformal prediction, uncertainty quantification, SHAP, XGBoost, explainable machine learning.

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List of Abbreviations

Abbreviation	Meaning
API	Application Programming Interface
CQR	Conformalized Quantile Regression
CSS	Cascading Style Sheets
CV	Cross-Validation
HTML	HyperText Markup Language
JS	JavaScript
MAE	Mean Absolute Error
MAPE	Mean Absolute Percentage Error
MAPIE	Model Agnostic Prediction Interval Estimator
RMSE	Root Mean Squared Error
SHAP	SHapley Additive exPlanations
XGBoost	eXtreme Gradient Boosting

Chapter 1

Introduction

Buying a house is the largest single purchase most families make, and almost nobody knows what the house they are looking at is actually worth. The usual recourse is an agent's opinion or a look at nearby recent sales; software that estimates a price automatically has existed for years and works reasonably well *on average*. That average is precisely the problem: it hides how the error is distributed across individual houses. Zillow ran the best-known such system in the United States and reported a typical error of around 7% for houses not currently listed for sale. Seven per cent sounds small until it is applied to one property, where it becomes a gap of several lakh rupees. In 2021 Zillow used its own estimates to buy houses at scale, then shut that business down after writing off \$304 million in a single quarter; its chief executive stated plainly that prices had proved harder to forecast than expected. The estimates were not useless – they were presented as more certain than they were.

Most house-price systems, commercial and academic alike, share this shape: train a model, report one error figure averaged over an entire test set, and stop. A user handed such an estimate has no way to judge whether their particular house is one the model handles well, no explanation of which details pushed the number up or down, and no warning when the property is unusual enough that the estimate should not be trusted at all. This project treats predicting the price as the easy part, and being honest about how much to trust each individual prediction as the part worth doing properly.

1.1 Problem Statement

Given the attributes of a residential flat in an Indian metro city – its floor area, locality, city, bedroom count, resale status, and a set of amenities – the problem this project addresses is threefold, not merely “predict the price”: (1) produce a point price estimate that is as accurate as the available data permits; (2) attach to that estimate a range that is verified, not assumed, to contain the true price at a stated rate; and (3) identify, before the fact, which predictions

are unreliable enough that the user should seek a professional opinion instead of trusting the model. A system that only solves the first of these three is solving the easier, and less useful, problem.

1.2 Objectives

1. Clean and prepare a real Indian metropolitan housing dataset, and train and compare a linear baseline, a Random Forest, and a gradient-boosted model (XGBoost), tuned by cross-validation and reported in currency terms rather than only as an abstract score.
2. Attach a 90% price range to every prediction using conformal prediction, and verify empirically – on data the model never trained or calibrated on – that a range claiming 90% reliability actually achieves it, reporting both the coverage obtained and the width of the range.
3. Explain each prediction by naming the features that pushed the estimate up or down relative to a typical house, so a user sees the reasoning behind a number, not just the number.
4. Detect houses that do not resemble the training data, or whose range comes out unusually wide, and flag those predictions as low-confidence with a recommendation to consult a professional valuer instead.
5. Build a working web application around the model and evaluate the whole system honestly, including separate results by price bracket and by city, so that a good overall average cannot hide poor behaviour on any one segment.

1.3 Contributions

1. A leakage-safe data pipeline for a real, messy Indian housing dataset: duplicate removal, principled handling of a placeholder “not recorded” code across 35 amenity columns, and features fit strictly on a training split so no information from calibration or test data leaks into preprocessing.
2. Price ranges built with MAPIE’s current (v1) conformal prediction API, using both a fixed-width split-conformal arm and an adaptive, quantile-regression-based (CQR) arm, empirically validated to reach approximately 90% coverage on a held-out test split touched exactly once.

3. Per-prediction explanations computed via XGBoost's native feature-contribution output, which is mathematically identical to SHAP's TreeExplainer values for tree ensembles, chosen specifically so the deployed web service does not need to import the heavier `shap` package at runtime.
4. A three-signal, empirically calibrated confidence flag (range width, novelty relative to training data, and missing important fields) that is shown in Chapter 4 to actually separate worse predictions from better ones, not merely to look reassuring.
5. A working, deployed system: a FastAPI backend serving a hand-built HTML/CSS/JavaScript frontend, plus a Streamlit application kept live as a fallback, both driven by the same trained pipeline with no duplicated modelling logic.

Chapter 2

Related Work / Literature Review

2.1 Tabular Model Choice: Trees versus Deep Learning

Grinsztajn, Oyallon and Varoquaux (NeurIPS, 2022) benchmarked deep learning against tree-based methods across 45 tabular datasets and found that tree-based models such as gradient boosting and random forests remain the strongest choice on medium-sized tables of around ten thousand rows, before even accounting for their much lower training cost. The Indian housing dataset used in this project sits squarely in that range, which is the reason this project does not use a neural network: on data of this shape and size, it would cost more and deliver less.

2.2 Hedonic Pricing Theory

Rosen (Journal of Political Economy, 1974) set out the economic idea underlying price prediction of this kind: a house is a bundle of separate attributes, and its price is built up from what buyers will pay for each of them. This is why explaining a prediction feature by feature is meaningful rather than decorative, and it is the basis for the per-house explanations produced in this project.

2.3 Conformal Prediction

Romano, Patterson and Candès (NeurIPS, 2019) and the survey by Angelopoulos and Bates (2021) describe conformal prediction, the main technique this project depends on for its price ranges. It turns any trained model into one that outputs a range instead of a single number, with a guarantee that the range contains the true value at the rate a user asks for. The guarantee assumes nothing about the shape of the data or the model, only that new houses resemble the ones used for calibration. The variant proposed by Romano and colleagues, used here as the

adaptive arm, widens the range where the model is genuinely uncertain and keeps it narrow for ordinary houses, which is exactly the behaviour a buyer needs.

2.4 Explainability

Lundberg and Lee (NIPS, 2017) introduced SHAP, which splits an individual prediction into the contribution of each input feature using a fair-division result from game theory. For tree ensembles specifically, this project computes the same underlying values through XGBoost’s own contribution API (Chen and Guestrin, ACM SIGKDD, 2016) rather than through the separate `shap` package, since the two are equivalent for tree models and avoiding the extra dependency matters for the memory budget of the deployed service.

2.5 The Zillow Offers Case

Finally, the closure of Zillow Offers in 2021, examined in the Journal of Information Systems Education (2024), is the clearest practical evidence for the problem this project addresses: a well-built model with a respectable average error still caused large losses, because the business acted on individual estimates without accounting for how uncertain any one of them was.

2.6 Summary of Research Gaps

Read together, these five threads point to the same gap this project fills. Tree ensembles are established as the right model family for tabular data of this size (Grinsztajn et al.), hedonic theory justifies feature-level explanation as economically meaningful rather than cosmetic (Rosen), conformal prediction supplies a statistically guaranteed way to quantify uncertainty around a point estimate (Romano et al.; Angelopoulos and Bates), and SHAP-equivalent contributions make that explanation computable without extra runtime cost (Lundberg and Lee; Chen and Guestrin). What none of this literature does on its own is combine all three – accurate point prediction, verified uncertainty, and per-house explanation – into one deployed system and then check, on real held-out data, that the uncertainty estimates and the confidence flag actually correlate with where the model is wrong. The Zillow Offers case (Journal of Information Systems Education, 2024) is the standing real-world argument for why that combination, not just the point estimate, is the part worth building.

Chapter 3

Methodology (Work Done)

3.1 Data

The dataset is a public collection of residential listing prices across six Indian metro cities – Bangalore, Chennai, Delhi, Hyderabad, Kolkata, and Mumbai – drawn from Kaggle. Combined across the six per-city files it contains 32,963 rows and 40 columns before cleaning: a target price, floor area, bedroom count, a resale flag, a free-text locality, and 35 binary amenity and furnishing indicators (gym, swimming pool, security, furnishing items, and similar). Two limitations are stated openly rather than hidden: the price column is an asking price, not a confirmed sale price, and there is no transaction date, so the model reflects the period the data was collected in rather than tracking the market over time.

Figure 3.1 shows why price and floor area are treated specially during feature engineering (Section 3.3): both are heavily right-skewed even before any cleaning, with a raw skewness of 13.92 for price and 3.55 for area, so a small number of very expensive, very large listings would otherwise dominate a model trained on the raw scale.

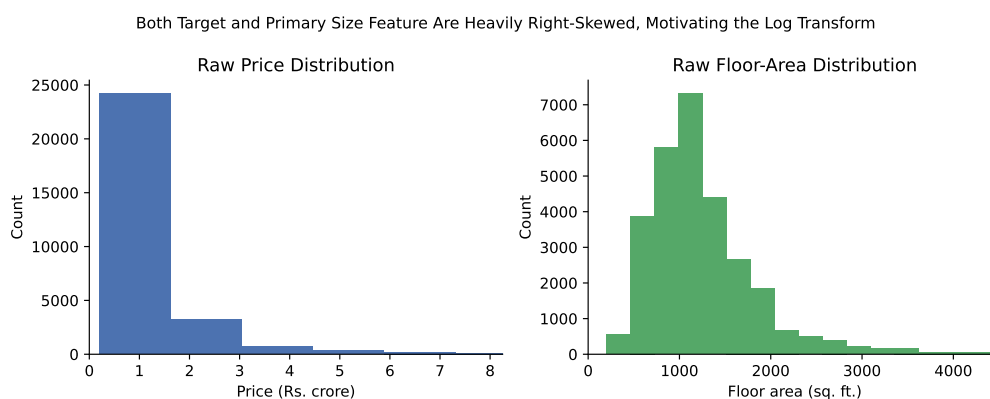


Figure 3.1: Distribution of raw listing price (in crore rupees) and floor area (in square feet) before cleaning, six-city combined data. Both distributions are strongly right-skewed, motivating the log-transform described in Section 3.3.

3.2 Cleaning

Schema inspection showed that each amenity column uses a placeholder value of 9 to mean “not recorded” for roughly 70% of rows; every value outside $\{0, 1\}$ is converted to a genuine missing value rather than assumed to be one specific guessed code. Exact duplicate rows are removed, dropping 3,828 of the 32,963 rows (11.6%) and leaving 29,135 rows for modelling. Rows with non-positive price or area are also dropped as data-entry errors.

3.3 Feature Engineering

The free-text locality column has a long tail of names occurring once or twice; localities occurring fewer than 15 times in the training split are collapsed into an “other” category before encoding, and the locality and city columns are then encoded with scikit-learn’s cross-fitted `TargetEncoder`, fit only on the training split so no calibration or test information leaks into the encoding. Price and floor area, both heavily right-skewed (Figure 3.1), are log-transformed. Price-per-area-style features are deliberately excluded, since they are computed from the target and would produce impressive but meaningless accuracy. No feature based on caste, religion, or community composition is used, a deliberate constraint given that locality names in Indian data can otherwise act as a proxy for such attributes.

3.4 Splitting, Training, and Uncertainty

The cleaned data is split 60/20/20 into training, calibration, and test portions. The calibration split is used only to build price ranges, never to fit or tune a model; the test split is touched exactly once, for final evaluation. Three models – regularised linear regression (Ridge), Random Forest, and XGBoost – are tuned by cross-validation on the training split alone (Table 4.1, Figure 3.2); XGBoost was selected as the point-prediction model. Price ranges at the 90% level are built two ways using MAPIE’s current (v1) API: split-conformal on the XGBoost model, and an adaptive, quantile-based variant (CQR) using a triplet of `HistGradientBoosting` quantile models, since MAPIE’s CQR implementation does not currently support XGBoost directly. Both arms are compared against a no-range baseline and a naive fixed-width band calibrated on the same calibration split, so the value of an adaptive range is demonstrated rather than assumed.

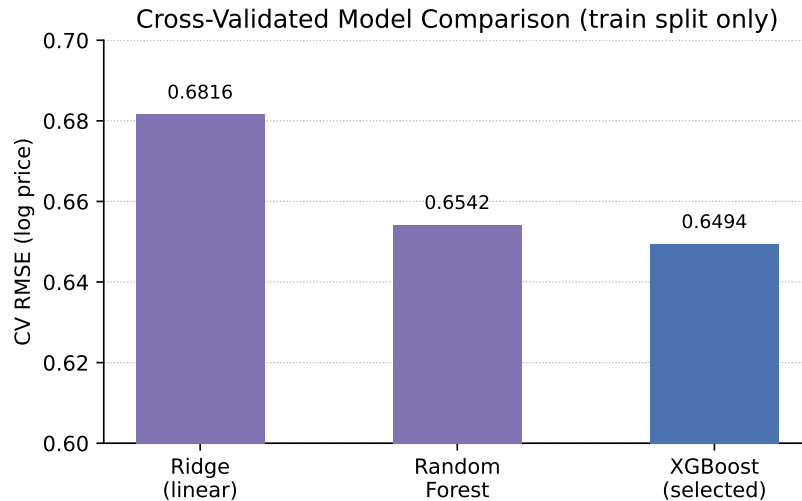


Figure 3.2: Cross-validated log-price RMSE for the three candidate models, training split (lower is better). XGBoost was selected.

3.5 Explanation and Confidence

Per-house explanations are produced from XGBoost’s native feature-contribution output rather than the `shap` package – the two are mathematically equivalent for tree ensembles, and avoiding `shap` keeps the deployed service’s memory footprint well within the 512 MB budget of its hosting tier. A confidence flag combines three signals computed on the calibration split: the width of a house’s price range relative to its predicted price, how unusual its features are relative to the training data (via an Isolation Forest fit on the training split only), and how many important fields were left unanswered. Each signal’s threshold is its own 90th percentile on the calibration split; a prediction is flagged when at least two of the three signals fire, or when any single signal is extreme (above its 97th percentile) – the second rule was added after finding that range width and novelty are, for this dataset, only weakly correlated, so requiring two ordinary signals to agree could under-flag genuinely unusual houses that only one signal catches strongly.

3.5.1 Confidence Flag Calibration

Figure 3.3 makes this weak correlation concrete: on the calibration split ($n = 5,827$), the Pearson correlation between a house’s novelty score and its range-width ratio is $r = -0.65$ – the two signals are, if anything, mildly anti-correlated rather than reinforcing one another. A house can therefore be highly novel relative to the training data while still receiving a comparatively narrow range (because the underlying quantile models happen to agree on it), or vice versa. This is precisely why the flagging rule combines three independent signals with a 2-of-3

vote rather than relying on any single one, and why the 97th-percentile “extreme single signal” override exists: without it, a house that is extremely novel but only moderately wide-ranged, or extremely wide-ranged but only moderately novel, could pass through both ordinary thresholds unflagged.

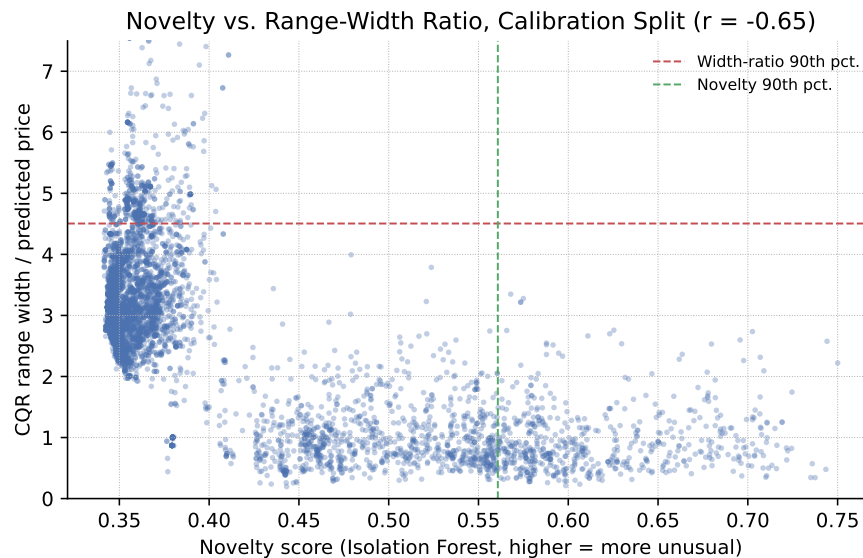


Figure 3.3: Novelty score versus range-width ratio on the calibration split ($n = 5,827$, $r = -0.65$), showing the two signals are not redundant with each other – justifying the three-signal, 2-of-3 design described in Section 3.5.

3.6 Deployment

3.6.1 System Architecture

The trained pipeline is served two ways from the same artefacts (Figure 3.4): a FastAPI backend (`/api/schema`, `/api/predict`, `/api/metrics`) behind a hand-built HTML, CSS, and JavaScript frontend, deployed on Render, and a Streamlit application kept live as a fallback on Streamlit Community Cloud. Both call the identical trained model, conformal objects, and confidence-flag logic through the shared `src/` package – `api/inference.py` reuses those modules directly rather than reimplementing any modelling step, so no ML logic is duplicated or can silently diverge between the two apps. The FastAPI service loads a trimmed `api_bundle.joblib` that omits the `shap` package as a runtime dependency, since XGBoost’s native contribution output is used instead (Section 3.5); this matters concretely on Render’s free tier, which caps the service at 512 MB of memory.

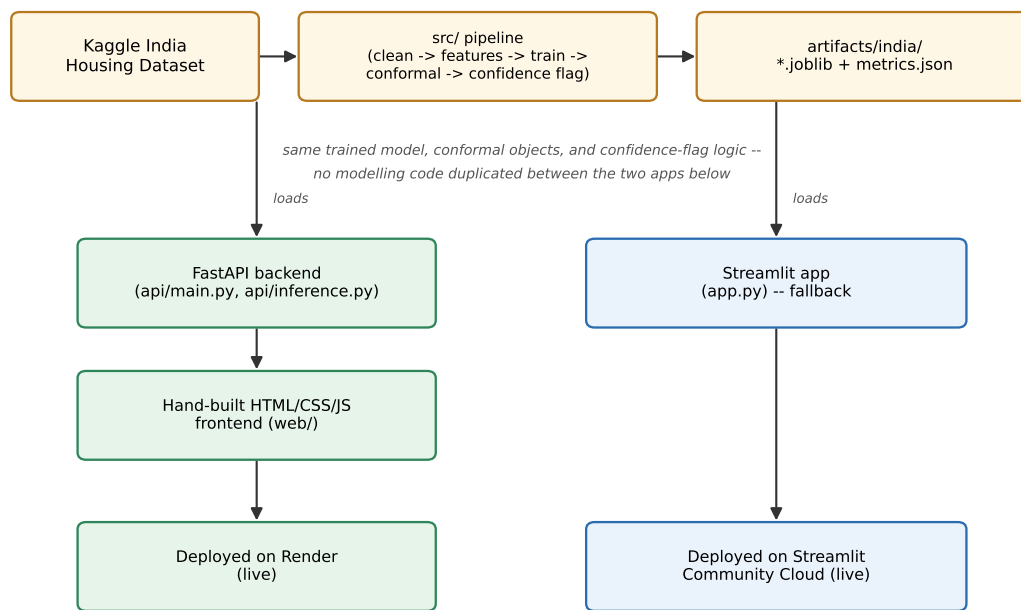


Figure 3.4: System architecture: two frontends served by one shared pipeline, with no modelling logic duplicated between them.

3.6.2 Web Form Design

The web form treats city and floor area as required, since a price estimate is least meaningful without either, while locality, bedroom count, and resale status may be left unanswered, which is tracked by the confidence flag rather than silently ignored. Figure 3.5 shows the deployed form.

The screenshot shows the 'HomeEstimate India' web application. The main heading is 'What's this flat actually worth?'. Below the heading, a text block explains that the application provides a predicted price with a 90% confidence range, validated by conformal prediction, and a plain explanation of what drove the estimate for flats in Bangalore, Chennai, Delhi, Hyderabad, Kolkata & Mumbai. The model used is 'xgboost' with a test-set error of 51.1% MAPE and Log-RMSE of 0.656. The input form is titled 'Location & basics' and includes the following fields:

- City:** A dropdown menu with 'Bangalore' selected. It is marked as 'Required'.
- Location:** A text input field with 'Jalahalli' entered. It has a checkbox for 'Unknown / skip'.
- Area:** A text input field with '1135' entered. It is marked as 'Required'.
- Bedrooms:** A text input field with '2' entered. It has a checkbox for 'Unknown / skip'.

Figure 3.5: The deployed application's input form (house-price-estimator-india.onrender.com).

Chapter 4

Results and Analysis

All figures in this chapter are computed on the test split, which was touched exactly once after every modelling and calibration decision was finalised.

4.1 Point-Prediction Accuracy

Table 4.1 shows the cross-validated log-price RMSE used to select XGBoost as the point-prediction model on the training split. Table 4.2 shows its accuracy on the held-out test split, reported both on the log scale used for model selection and in currency terms that are directly interpretable.

Table 4.1: Cross-validated log-price RMSE on the training split (lower is better).

Model	CV RMSE (log price)
Ridge (linear baseline)	0.6816
Random Forest	0.6542
XGBoost (selected)	0.6494

Table 4.2: Point-prediction accuracy on the test split (XGBoost).

Metric	Value
RMSE (log price)	0.656
Mean Absolute Error	Rs. 60.21 lakh
RMSE	Rs. 2.02 crore
MAPE	51.1%

A MAPE of 51.1% is markedly weaker than a well-behaved regression problem, and it is reported here without averaging it away. It reflects the dataset's own stated limitations – asking prices rather than confirmed sales, no transaction date, and roughly 70% missing amenity data

– rather than a failure of the modelling approach; Figure 4.2 breaks this down further by price bracket.

4.2 Price-Range Coverage

Table 4.3 and Figure 4.1 compare all four range-building methods against their shared 90% target. Every method lands within about one percentage point of the target, which is itself a result: it shows the calibration procedure works, and that the adaptive (CQR) arm achieves the same reliability as the simpler alternatives while, as shown by its narrower median width, concentrating its width where it is actually needed rather than applying it uniformly.

Table 4.3: 90% price-range coverage and width by method, test split.

Method	Coverage	Avg. width	Median width
CQR (adaptive)	89.9%	Rs. 2.35 Cr	Rs. 2.02 Cr
Split-conformal (XGBoost)	90.5%	Rs. 2.19 Cr	Rs. 1.70 Cr
Split-conformal (HistGBR)	90.3%	Rs. 2.10 Cr	Rs. 1.64 Cr
Naive fixed-width baseline	90.1%	Rs. 2.18 Cr	Rs. 2.18 Cr

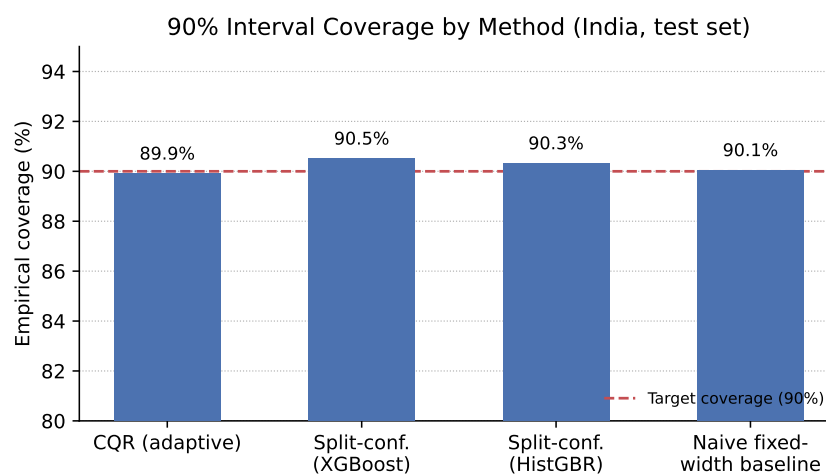


Figure 4.1: Empirical 90% coverage achieved by each range-construction method on the test split, against the 90% target (dashed line).

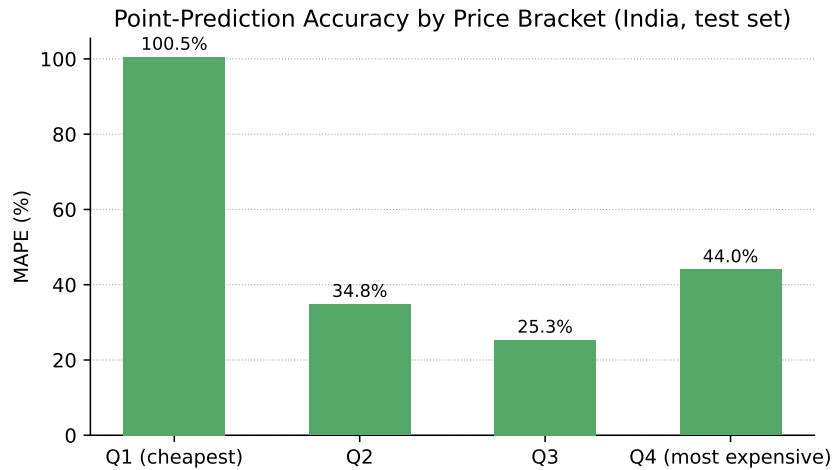


Figure 4.2: Point-prediction MAPE by price quartile, test split. The cheapest bracket (Q1) is proportionally hardest, consistent with asking-price noise mattering more for lower-value listings.

4.3 Coverage by City

No city in the test set was excluded from this breakdown, since every one had at least the minimum 100 test rows required to report a reliable per-segment number (Table 4.4, Figure 4.3). Coverage stays close to 90% in every city, though average range width varies by roughly a factor of three between the cheapest and most expensive markets, showing that the adaptive range genuinely tracks each city’s own price volatility rather than applying one national-average width everywhere.

Table 4.4: CQR coverage and average width by city, test split (all $n \geq 100$).

City	n (test)	Coverage	Avg. width
Bangalore	1,107	91.9%	Rs. 2.02 Cr
Chennai	892	90.2%	Rs. 1.56 Cr
Delhi	842	89.0%	Rs. 3.31 Cr
Hyderabad	395	88.9%	Rs. 0.92 Cr
Kolkata	1,260	89.2%	Rs. 1.97 Cr
Mumbai	1,331	89.7%	Rs. 3.33 Cr

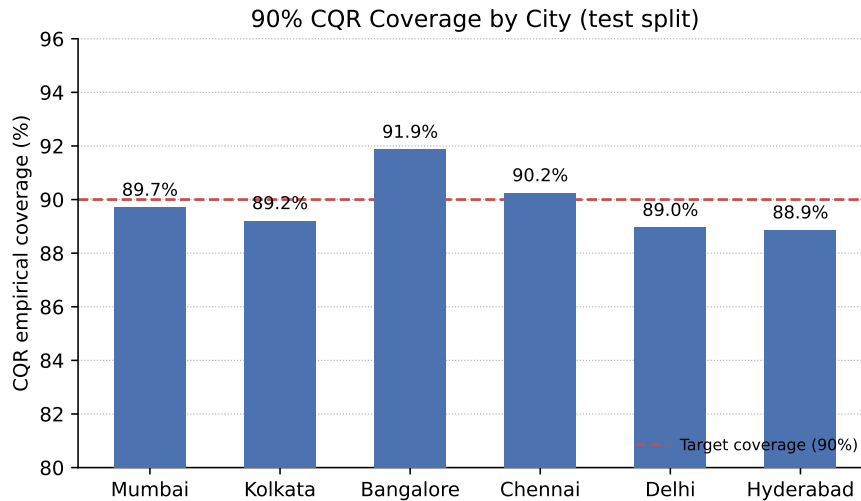


Figure 4.3: CQR coverage by city, test split, against the 90% target (dashed line). All six cities land within roughly two points of target.

4.4 Does the Confidence Flag Mean Anything

The confidence flag is only useful if a flagged prediction is actually worse; Table 4.5 confirms this on the test split, which the flag’s calibration never saw. Flagged predictions show 62.1% MAPE against 50.4% for the rest – a real, sizeable gap, and evidence that the flag is catching genuine weaknesses in the model rather than adding noise.

Table 4.5: Confidence-flag validation, test split.

Group	n	MAE	MAPE
Flagged low-confidence	310	Rs. 83.64 lakh	62.1%
Not flagged	5,517	Rs. 58.90 lakh	50.4%

4.5 The Deployed Application in Use

Figure 4.4 shows a live prediction on the deployed application for a 1,450 sq. ft. resale flat in Whitefield, Bangalore: a predicted price of Rs. 73.99 lakh with a 90% range of Rs. 27.57 lakh to Rs. 2.97 crore, rendered with the same range-visualisation logic evaluated quantitatively in Section 3.3–4 above. This end-to-end trace, from a raw form submission through the deployed API back to a rendered chart, is what Section 1.2’s fifth objective – a working web application, evaluated honestly – refers to in practice, not only as a set of offline numbers.

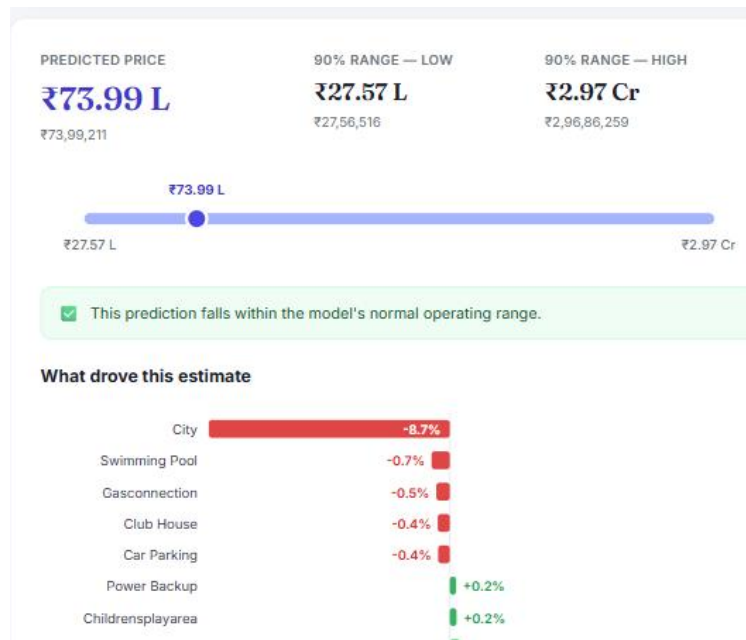


Figure 4.4: A live prediction on the deployed application, showing the predicted price, the full 90% range, the low-confidence check, and most of the per-feature explanation chart for a sample Bangalore flat.

The application also exposes its own validation numbers directly to the user, rather than asking them to trust the model blindly: an in-app “Model performance & validation” panel (Figure 4.5) surfaces the same point-accuracy and coverage-by-method figures reported in Tables 4.2 and 4.3, computed once at deploy time and served directly from `/api/metrics` rather than recomputed per request.

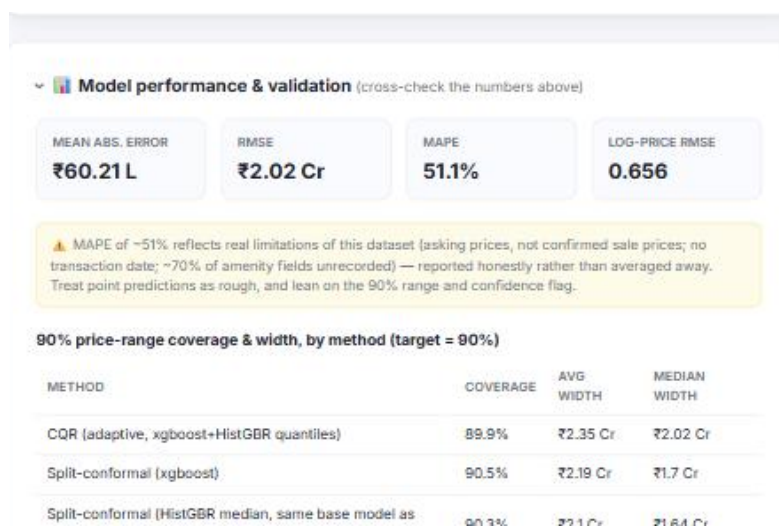


Figure 4.5: The deployed application’s “Model performance & validation” panel, showing the point-accuracy metrics from Table 4.2 and the coverage-by-method table from Table 4.3, surfaced directly to the user rather than left in an offline report only.

Chapter 5

Conclusion and Future Work

All five objectives set out in Section 1.2 were met on real data, not only in principle. Three candidate models were cleaned, trained, and compared honestly rather than only reporting the winner; price ranges were built with conformal prediction and verified, not assumed, to reach their claimed 90% coverage on data touched exactly once; every prediction carries a feature-level explanation and a confidence flag shown to correlate with real prediction error; and the whole system runs as a deployed web application rather than a notebook. The project’s central finding is that reliability can be measured and delivered even when point accuracy is weak: a 51.1% MAPE, driven by asking-price noise and missing data rather than a modelling shortfall, still comes with price ranges that hit their 90% target and a confidence flag that reliably identifies the predictions least worth trusting.

Table 5.1 traces each objective from Section 1.2 to the chapter and section where it was addressed and verified, so the correspondence between what was promised and what was delivered is explicit rather than left to be inferred.

Table 5.1: Objectives traceability.

#	Objective	Addressed in
1	Clean data; train and compare Ridge, Random Forest, XGBoost	Sections 3.1–3.4, 4.1
2	90% conformal price ranges, empirically verified	Sections 3.4, 4.2
3	Per-house feature explanations	Section 3.5
4	Low-confidence flag for unusual/wide-range houses	Sections 3.5, 4.4
5	Working web application, evaluated honestly	Sections 3.6, 4.3, 4.5

The clearest limitation is the data itself. Asking prices are not confirmed sales, there is no transaction date to track the market over time, and roughly 70% of amenity information is

unrecorded. None of this can be fixed by a better model; it requires a better data source. Future work should prioritise, in order: (1) a verified-transaction dataset with sale dates, which would likely improve point accuracy far more than further model tuning; (2) extending coverage to more Indian cities and periodically retraining as prices move, since the current model reflects only the period its training data was collected in; (3) a native mobile client against the existing FastAPI backend, which already separates the model-serving API from the frontend; and (4) revisiting the confidence flag's two-tier design as more data becomes available, since the anti-correlation found between range width and novelty for this dataset (Section 3.5) is itself worth further investigation rather than only working around.

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Appendix A

Project Directory Structure

The repository is organised so that no modelling logic is duplicated between the two deployed applications; both call directly into `src/`.

```
src/
|-- config.py           # paths, seed, per-market registry
|-- data/              # acquire, clean, split
|-- features/          # per-market feature engineering + shared preprocessing
|-- models/            # train/tune, quantile models, persistence
|-- uncertainty/       # conformal prediction (MAPIE) + baselines
|-- explain/
|   |-- feature_names.py # shap-independent column-name aggregation (shared)
|   |-- shap_explainer.py # SHAP explanations (pipeline/Streamlit only)
|-- confidence/        # low-confidence flag
|-- evaluation/        # metrics, coverage/width tables, report writer
'-- pipeline.py        # end-to-end orchestration
api/
|-- main.py            # FastAPI app: /api/schema, /api/predict, /api/metrics
'-- inference.py       # request handling, reuses src/ (no ML logic duplicated)
web/                  # static HTML/CSS/JS frontend served by api/main.py
app.py                # Streamlit app (kept live as a fallback)
tests/                # unit tests
reports/              # generated evaluation reports
artifacts/{ames,india}/ # model_bundle.joblib (full) + api_bundle.joblib (trimmed)
render.yaml           # Render.com Blueprint for the FastAPI service
requirements-api.txt  # slim deps for the deployed API (no shap/streamlit/kaggle)
```